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Inventor(s)	Kennedy; Ryan J. et al.

Smart circuit system for an electric vehicle charging system

Abstract

A smart circuit system configured to enhance circuit protection by directly interfacing with a circuit breaker is disclosed. The system comprises a digital or wireless interface, a ground fault circuit interrupter (GFCI), a load control device, electric vehicle supply equipment (EVSE) functionality, metering functionality, a diagnosis module, and a communication module. The system can be integrated within an existing panel or any other system requiring load control or EVSE functionality. The GFCI function can be externally added to any circuit breaker. The diagnosis module assesses circuit breaker health by comparing input voltage and voltage trends with voltages measured by other smart circuits within the same panel. The communication module establishes a network among smart circuits for data exchange, enabling local calculations and independent decision-making. The system also includes a load monitoring module for assessing load type and health.

Inventors: **Kennedy; Ryan J. (Cornelius, NC), Harris; Michael J. (Concord, NC)**

Applicant: **Atom Power, Inc.** (Huntersville, NC)

Family ID: **86898080**

Assignee: **Atom Power, Inc. (Huntersville, NC)**

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Primary Examiner: Ortiz; Elim

Attorney, Agent or Firm: Withrow & Terranova, PLLC

Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of provisional patent application Ser. No. 63/346,410, filed May 27, 2022, the disclosure of which is hereby incorporated herein by reference in its entirety. (2) This application is a continuation-in-part of U.S. patent application Ser. No. 18/178,024, filed Mar. 3, 2023, which is a continuation-in-part of U.S. patent application Ser. No. 17/543,893, filed Dec. 7, 2021, now U.S. Pat. No. 11,884,177, which claims the benefit of provisional patent application Ser. No. 63/122,681, filed Dec. 8, 2020, the disclosures of which are hereby incorporated herein by reference in their entireties.

FIELD OF THE DISCLOSURE
(1) A smart circuit system designed to enhance circuit protection within an electric vehicle charging system.
BACKGROUND
(2) Electric vehicle (EV) charging systems are becoming increasingly prevalent in modern society as the transition away from fossil-fuel-powered vehicles toward plug-in electric vehicles (PEVs) continues. Despite many advancements in PEV design in recent years, electric vehicle supply equipment (EVSE) in state-of-the-art EV charging systems remains relatively primitive. To compensate for the limited capabilities, modern EV charging systems tend to be built as “one-size-fits-all” systems. Unfortunately, this one-size-fits-all approach is expensive to implement since it requires the use of oversized electrical power distribution equipment (e.g., heavy gauge cables and wires) in order to accommodate most every conceivable PEV charging need. State-of-the art EVSE also fails to address several safety concerns that are potentially hazardous to EV charging system users. The present disclosure addresses these problems, providing an EV charging system that is not only safer to use and operate but that also maximizes infrastructure usage and avoids the high cost and need for oversized power distribution equipment.
SUMMARY
(3) A smart circuit system configured to enhance circuit protection by directly interfacing with a circuit breaker is disclosed. The system comprises a digital or wireless interface, a ground fault circuit interrupter (GFCI), a load control device, electric vehicle supply equipment (EVSE) functionality, metering functionality, a diagnosis module, and a communication module. The system can be integrated within an existing panel or any other system requiring load control or EVSE functionality. The GFCI function can be externally added to any circuit breaker. The diagnosis module assesses circuit breaker health by comparing input voltage and voltage trends with voltages measured by other smart circuits within the same panel. The communication module establishes a network among smart circuits for data exchange, enabling local calculations and independent decision-making. The system also includes a load monitoring module for assessing load type and health.
(4) In some embodiments, the smart circuit system is integrated with an electric vehicle (EV) charging system includes a plurality of electrical vehicle supply equipment (EVSE) units, a plurality of associated EV charging stations, phase conductors coupled between the charging stations and corresponding bi-directional solid-state switches and a neutral conductor configured to complete charging circuits with associated ones of the phase conductors, and an EVSE communications bus. Each EVSE unit includes a microcontroller unit (MCU) and driver that control current flow through the bi-directional solid-state switches providing charging current to respective EV charging stations and connected plug-in EVs (PEVs). The MCUs communicate over the EVSE communications bus and, as PEVs plug into, charge, and unplug from the plurality of EV charging stations, reallocate or reapportion an available supply current among the plurality of EVSE units while also dynamically adjusting one or more circuit protection attributes provided by

the EVSE units, including, for example, the continuous current rating(s) of one or more EVSE units.
(5) Further features and advantages of the present disclosure, including a detailed description of the above-summarized and other exemplary embodiments of the present disclosure, will now be described in detail with respect to the accompanying drawings, in which like reference numbers are used to indicate identical or functionally similar elements.
(6) In another aspect, any of the foregoing aspects individually or together, and/or various separate aspects and features as described herein, may be combined for additional advantage. Any of the various features and elements as disclosed herein may be combined with one or more other disclosed features and elements unless indicated to the contrary herein.
(7) Those skilled in the art will appreciate the scope of the present disclosure and realize additional aspects thereof after reading the following detailed description of the preferred embodiments in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES
(1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure and, together with the description, serve to explain the principles of the disclosure.
(2) FIG. **1** is a block diagram of an electric vehicle (EV) charging system, according to an embodiment of the present disclosure.
(3) FIG. **2** is a drawing illustrating the salient components of an EV supply equipment unit (EVSE) or “EVSEP,” according to an embodiment of the present disclosure.
(4) FIG. **3** is a drawing depicting an exemplary time-current characteristic (TCC) curve according to which an EVSEP can be configured and controlled to operate, including various dynamically adjustable circuit protection attributes.
(5) FIG. **4** is a drawing depicting the principal components of a three-phase embodiment of an EVSEP, according to an embodiment of the present disclosure.
(6) FIG. **5**A is a drawing depicting how the bi-directional solid-state switch and air gap disconnect unit of an EVSEP are configured when the EVSEP is in an “ON” state.
(7) FIG. **5**B is a drawing depicting how the bi-directional solid-state switch and air gap disconnect unit of an EVSEP are configured when the EVSEP is in an “OFF” state.
(8) FIG. **5**C is a drawing depicting how the bi-directional solid-state switch and air gap disconnect unit of an EVSEP are configured when the EVSEP is in a “STANDBY” state.
(9) FIG. **6**A is a face view drawing of a connector portion of an alternating current (AC) charging cable handle that can be employed at a charging station in the EV charging system depicted in FIG. **1**, according to one embodiment of the present disclosure.
(10) FIG. **6**B is a perspective view drawing of an AC charging cable handle that can be employed at a charging station in the EV charging system depicted in FIG. **1**, according to one embodiment of the present disclosure.
(11) FIG. **7** is a face view drawing of a connector portion of an AC and fast direct current (DC) “combo” connector that can be employed in the EV charging system depicted in FIG. **1**, according to one embodiment of the present disclosure.
(12) FIG. **8**A is a first portion of a flowchart of a charge control and circuit protection method that an EVSEP performs, according to an embodiment of the present disclosure.
(13) FIG. **8**B is a second portion of a flowchart of a charge control and circuit protection method that an EVSEP performs, according to an embodiment of the present disclosure.
(14) FIG. **8**C is a third portion of a flowchart of a charge control and circuit protection method that

an EVSEP performs, according to an embodiment of the present disclosure.
(15) FIG. **8**D is a fourth portion of a flowchart of a charge control and circuit protection method that an EVSEP performs, according to an embodiment of the present disclosure.
(16) FIG. **9** is a chart highlighting various charge control and circuit protection attributes involved in a weighting-based charge control and circuit protection method, according to one embodiment of the present disclosure.
(17) FIG. **10** is a chart including various charge control and circuit protection attributes involved in a priority-dependent weighting-based charge control and circuit protection method, according to one embodiment of the present disclosure.
(18) FIG. **11** is a drawing illustrating how an EVSEP and associated EV charging station are configured in relation to a plug-in EV (PEV) that supports AC charging, according to one embodiment of the present disclosure.
(19) FIG. **12** is a drawing illustrating how a modified EVSEP, designed and constructed to support DC-fast charging, and associated EV charging station are configured in relation to a PEV that supports DC-fast charging, according to one embodiment of the present disclosure.
(20) FIG. **13** is a drawing depicting an EV charging system that is adapted and well-suited to serve as a fleet charging facility for EV buses and other municipal EVs, according to an embodiment of the present disclosure.
(21) FIG. **14** is a drawing of an EVSEP that includes an internal ground fault current transformer (GFCT) that is configured to detect ground fault currents in two-phase conductors and a neutral conductor, according to an embodiment of the present disclosure.
(22) FIG. **15** is a drawing of a version the EVSEP that is configured to respond to a ground fault sense signal generated by an external version of the GFCT that is configured to detect ground fault currents in two-phase conductors and a neutral conductor, according to an embodiment of the present disclosure.
(23) FIG. **16** is a drawing of an EVSEP that includes a version of the internal GFCT that is configured to detect ground fault currents in three-phase conductors and a neutral conductor, according to an embodiment of the present disclosure.
(24) FIG. **17** is a drawing of an EVSEP that is configured to respond to a ground fault sense signal generated by an external version of the GFCT that is configured to detect ground fault currents in three-phase conductors and a neutral conductor, according to an embodiment of the present disclosure.
(25) FIG. **18** is a drawing of a version of the electric vehicle charging system that is configured to charge two electric vehicles per EVSEP having the internal GFCT as depicted in FIG. **14** and further including a gateway and pilot module configured to communicate with pedestals, according to an embodiment of the present disclosure.
(26) FIG. **19** is a drawing of a version of the electric vehicle charging system that is configured to charge two electric vehicles per EVSEP and includes a gateway and a substantially low-voltage DC power and data module such as the power over ethernet type to communicate with smart pilot modules in pedestals, according to an embodiment of the present disclosure.
(27) FIG. **20** is a drawing of a version of the electric vehicle charging system that is configured to charge two electric vehicles per EVSEP with one EVSP configured to respond to ground fault sense signals generated by external GFCTs located at pedestals, according to an embodiment of the present disclosure.
(28) FIG. **21** is a drawing of a version of the electric vehicle charging system that is configured to charge three electric vehicles per EVSEP and includes the gateway and the substantially low-voltage DC power and data module to communicate with smart pilot modules in pedestals, according to an embodiment of the present disclosure.
(29) FIG. **22** is a drawing of a version of the electric vehicle charging system that is configured to charge three electric vehicles per EVSEP with one EVSP configured to respond to ground fault

sense signals generated by external GFCTs located at pedestals, according to an embodiment of the present disclosure.

(30) FIG. **23** is a drawing of an electromechanical version the EVSEP that is configured to respond to a ground fault sense signal generated by an external version of the GFCT that is configured to detect ground fault currents in two-phase conductors.

(31) FIG. **24** is a drawing of an electromechanical version of the electric vehicle charging system that is configured to charge two electric vehicles per EVSEP including the gateway and pilot module configured to communicate with pedestals, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(32) The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

(33) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(34) It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

(35) Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

(36) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(37) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as

having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.
(38) Embodiments are described herein with reference to schematic illustrations of embodiments of the disclosure. As such, the actual dimensions of the layers and elements can be different, and variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are expected. For example, a region illustrated or described as square or rectangular can have rounded or curved features, and regions shown as straight lines may have some irregularity. Thus, the regions illustrated in the figures are schematic and their shapes are not intended to illustrate the precise shape of a region of a device and are not intended to limit the scope of the disclosure. Additionally, sizes of structures or regions may be exaggerated relative to other structures or regions for illustrative purposes and, thus, are provided to illustrate the general structures of the present subject matter and may or may not be drawn to scale. Common elements between figures may be shown herein with common element numbers and may not be subsequently re-described.
(39) Referring to FIG. **1**, there is shown an electric vehicle (EV) charging system **100**, according to one embodiment of the present disclosure. The electric vehicle charging system **100** comprises a plurality of Electric Vehicle Supply Equipment units (also referred to as "EV supply equipment units" and "EVSEPs **102**" in the detailed description that follows) electrically connected and configured within an EVSEP electrical distribution panel **104** (referred to as the "EVSEP panel **104**" below). The electric vehicle charging system **100** further comprises electrical power distribution cables **106**, junction boxes ("J-boxes") **108** located in the general vicinity of a plurality of associated charging stations **110**, and plug-in charging cables **112**. The electrical power distribution cables **106** distribute electrical power from the various EVSEPs **102** to the plurality of J-boxes **108** and associated charging stations **110**, thereby providing a plurality of charging current sources for plug-in electric vehicles (PEVs) that connect to (i.e., "plug into") the charging stations **110**.
(40) Each EVSEP **102** of the EV charging system **100** serves two primary purposes. First, each EVSEP **102** controls the maximum allowable charging current its associated charging station **110** is allowed to supply to a connected PEV at any given time. Second, each EVSEP **102** provides circuit protection to its associated charging station **110** and connected PEV, based on the maximum allowable charging current. The circuit protection and charge control functions provided by the EVSEPs **102** are self-operating, i.e., require no human interaction, and are both dynamically adjustable, individually and collectively, in other words, are adjusted in real time as PEVs connect to and from the charging stations **110**. Prior art electric vehicle charging systems, in contrast, employ primitive vehicle supply equipment (EVSE) that lack these capabilities. Consequently, if an EV charging system facility provides, say, ten charging stations at 50 A each, the power distribution infrastructure for the facility must be constructed so that all charging stations are capable of supplying the full 50 A, all at the same time, even though all EVSEPs using the full 50 A all at the same time is unlikely or rarely to occur. In other words, the EV charging system facility must be constructed using oversized components, including heavy gauge wires and cables, and must include line-side power distribution infrastructure that is capable of, at any one time, sinking 500 A from the utility grid. Not only is the oversized infrastructure expensive to implement, it results in safety concerns, particularly in circumstances where consumers with PEVs having low-capacity battery packs, in need of relatively low charging currents, must connect to and charge from a high-current/high-voltage charging station. Conventional EVSE also does not typically provide any circuit protection to its charging stations and connected PEVs, relying instead entirely on the PEV's battery management system for circuit protection and/or on electromechanical circuit breakers located external to the EV charging system to provide the circuit protection. Any circuit protection that is provided is static, i.e., is not dynamically adjustable. The EV charging system **100** of the

present disclosure avoids these problems, not only because the EVSEPs **102** have built-in circuit protection capabilities but also by virtue of the fact that the circuit protection and charge control functions provided by the EVSEPs **102** are dynamically adjustable, both individually and collectively. To emphasize these attributes and the various other advantages the EVSEPs **102** of the present disclosure have over conventional EVSEs, the letter "P" is appended to the acronym "EVSE" (i.e., EVSEP **102**) in the description of the exemplary embodiments of the present disclosure described herein—the "P" signifying the "Plus" capabilities the EVSEPs **102** have compared to conventional EVSEs, including the EVSEPs' dynamically adjustable circuit protection and dynamically adjustable charge control and charge allocation capabilities.

(41) The EVSEP panel **104** is configured to receive electrical power from the utility grid (e.g., at a utility service drop) or, alternatively, from another upstream EVSEP panel or upstream conventional panel. In accordance with one embodiment of the present disclosure, the EVSEP panel **104** further includes an EVSEP communications bus **116** (for example, a controller area network (CAN) bus or an inter-integrated circuit (I.sup.2C) bus) and a gateway **118** that is communicatively coupled to the EVSEPs **102**. The MCUs **202** (see FIG. **2**) of the EVSEPs **102** are configured and programmed to communicate with one another over the EVSEP communications bus **116** and, among other things, to assess, compute and, when necessary, apportion the total supply current supplied by EVSEP panel **104** among the various charging stations **110**. The gateway **118** serves to distribute control pilot signals from the EVSEPs **102** to handles **114** of the charging cables **112** located at the charging stations **110**, e.g., via the J-boxes **108**, and the control pilot signals are used to negotiate the charging currents supplied from the charging stations **110** to their connected PEVs, as will be described in more detail below.

(42) FIG. **2** is a block diagram of one of the EVSEPs **102** employed in the EV charging system **100**, highlighting its principal components. The principal components include: a microcontroller unit (MCU) **202**; a bi-directional solid-state switch **204**; a driver circuit **206**, which controls whether the bi-directional solid-state switch **204** is ON (closed) or OFF (open); an air gap disconnect unit **208** that provides a physical separation between line and load, otherwise known as disconnect or lockable disconnect; Hall Effect sensors **210** or other current-sensing technology; computer readable memory (CRM) **212**; and a ground-fault circuit interrupter (GFCI) **214**. The bi-directional solid-state switch **204** comprises one or more power semiconductor devices connected in series with a closeable air gap **216** of the air gap disconnect unit **208**, between input (line-side) terminals **218** and output (load-side) terminals **220**. The input terminals **218** are electrically connected to hot bus bar, neutral, and ground connections in the EVSEP panel **104**, and the output terminals **220** are electrically connected to electrical power distribution cables or wires **106**, which, as explained above, serve to distribute electrical power from each EVSEP **102** to its associated J-Box **108** and charging station **110**. In a preferred embodiment of the present disclosure the one or more bi-directional power semiconductors forming the bi-directional solid-state switch **204** comprise(s) one or more or more silicon carbide (SiC) metal-oxide-semiconductor field-effect transistors (MOSFETs). However, other types of power semiconductor devices can be used, as will be appreciated by those of ordinary skill in the art.

(43) The CRM **212** in each of the EVSEPs **102** comprises flash memory and/or electrically erasable programmable read-only memory (EEPROM) for storing the computer program instructions, and random-access memory (RAM), which the EVSEP's MCU **202** uses to perform the various operations specified by the computer program instructions. The CRM **212** may be entirely external to the MCU **202** (as depicted in FIG. **2**), embedded entirely within the MCU **202**, or may be partly external to the MCU **202** and partly integrated within the MCU **202**, as will be appreciated and understood by those of ordinary skill in the art. Among other things, the computer program instructions include instructions that provide the MCU **202** in each EVSEP **102** the ability to set, control, and adjust the circuit protection settings of provided by the EVSEP **102** and instructions that provide the MCUs **202** the ability to set, control, and adjust how much charging current is

allowed to flow through the EVSEP **102** to its associated charging station **110**.

(44) Under the control of its MCU **202** the bi-directional solid-state switch **204** in each EVSEP **102** determines whether charging current is allowed to flow to the EVSEP's associated charging station **110**. As explained in further detail below, when the bi-directional solid-state switch **204** in a given EVSEP **102** is switched ON and the EVSEP's air gap disconnect unit **208** is closed (in embodiments of the EVSEP that utilize an air gap disconnect unit **208**), charging current is able to flow through the solid-state switch **204** and closed air gap **216** of the air gap disconnect unit **208**, to the EVSEP's associated charging station **110**, and ultimately to a PEV connected to (i.e., plugged into) the charging station **110**, via associated J-box **108** and charging cable **112**. However, upon the EVSEP's MCU **202** determining, with the aid of measurements taken by its Hall effect sensors **210**, that a short circuit is present in the EVSEP's **102**'s load circuit, the MCU **202** directs the driver circuit **206** in the EVSEP **102** to switch the bi-directional solid-state switch **204** OFF, as soon as it is possible. It also generates a solenoid trigger signal that triggers a solenoid in the air gap disconnect unit **208** to open the air gap **216** and thereby isolate the charging station **110** and PEV and prevent any additional current from flowing to the associated charging station **110**. By employing solid-state switches **204** in the EVSEPs **102**, the EVSEPs **102** are able to isolate short circuits over a thousand times faster than conventional electromechanical circuit breakers. There are various ways in which the air gap disconnect unit **208** in the EVSEPs **102** can be designed to trigger and various ways in which EVSEPs' bi-directional solid-state switches **204** can be controlled to switch ON and OFF (e.g., entirely hardware controlled or hardware and software controlled). Some examples of how the EVSEPs **102** may be adapted to perform these functions can be found in commonly owned U.S. Pat. No. 10,541,530 and co-pending and commonly assigned U.S. patent application Ser. No. 16/898,538, now U.S. Pat. No. 11,322,916, both of which are incorporated herein by reference.

(45) In one embodiment of the present disclosure the MCUs **202** and bi-directional solid-state switches **204** in the EVSEPs **102** not only provide short-circuit protection to the charging stations **100** and connected PEVs but the MCU **202** and bi-directional solid-state switch **204** in each EVSEP **102** also provides overcurrent protection and is configured and controlled by its associated MCU **202** to operate according to a dynamically adjustable time-current characteristic (TCC) curve, for example, as illustrated in FIG. **3**. The computer program instructions (e.g., firmware) stored in the CRM **212** of each EVSEP **102** provides the MCU **202** the ability to set, control, and adjust any one or more of the various circuit protection attributes of the TCC curves, including, but not limited to: the EVSEP's continuous current rating Ir, long-time delay and short-time delay trip settings, short time pickup current, and instantaneous pickup current. In a preferred embodiment of the present disclosure the computer program instructions further include instructions that provide the MCU **202** in each EVSEP **102** the ability to communicate in real time with the MCUs **202** of other EVSEPs **102** in the EVSEP panel **104**, over the EVSEP communications bus **116**, and instructions that provide the MCU **202** in a given EVSEP **102** the ability to adjust the TCC curve of its respective bi-directional solid-state switch **204**, dynamically and in real time, depending on the magnitudes of the currents that are currently allocated to other charging stations **110** and associated PEVs via other EVSEPs **102** in the EV charging system **100** (or depending on the magnitudes of the maximum allowable charging currents I_CHARGE currently set by the EVSEPs **102**) and/or depending on changes (increases or decreases) in the magnitudes of the allocated currents (or depending on changes (increases or decreases) in the magnitudes of the maximum allowable charging currents I_CHARGE).

(46) The EVSEP **102** depicted in FIG. **2** can be easily modified to be a three-phase EVSEP, for use in a three-phase EV charging system. As illustrated in FIG. **4**, the three-phase embodiment of the EVSEP **102** has three current carrying conductors (L1, L2, L3), instead of just two L1 and L2, and a three-phase bi-directional solid-state switch. Although not shown in the drawing, the three-phase embodiment of the EVSEP also includes, preferably though not necessarily, a modified air gap

disconnect unit with a closeable air gap in all three lines L1, L2, and L3.
(47) It should be mentioned that whereas the exemplary embodiment of the EVSEP **102** described herein and depicted in FIG. **2** and other drawings of this disclosure includes air gap disconnect unit **208**, in other embodiments of the present disclosure the EVSEP **102** does not include an air gap disconnect unit but rather relies solely on the bi-directional solid-state switch **204** to provide fault and overcurrent protection. Including the air gap disconnect unit **208** is preferable in most circumstances and applications, however, since it isolates the EVSEP's **102**'s charging station **110** and associated PEV from the rest of the EV charging system **100** when a fault or other potentially hazardous situation occurs.
(48) At any given time during operation of the EV charging system **100**, each EVSEP **102** in the EVSEP panel **104** is configured in one of three possible states: an “ON” state, an “OFF” state, or a “STANDBY” state. These three states are illustrated in FIGS. **5**A-**5**C. When in the “ON” state (FIG. **5**A), the air gap disconnect unit **208** is configured so that the air gap **216** is closed and the bi-directional solid-state switch **204** is ON. It is only in this “ON” state that the EVSEP **102** allows charging current to flow to its associated charging station **110** and connected PEV. During charging, the charging current supplied from the EVSEP panel **104** flows through the associated EVSEP **102** (i.e., through the closed air gap **216** of the air gap disconnect unit and the ON bi-directional solid-state switch **204**), through corresponding electrical power distribution cables **106**, charging cable **112**, J-Box **108**, and charging station **110**, and finally to the battery charger located in the PEV connected to the associated charging station **110**. Each EVSEP **102** in the EVSEP panel **104** is configured to protect an associated load circuit containing the charging station **110** and plugged-in PEV, and will transition to the OFF state if the EVSEP **102** detects a short circuit in the electrical power distribution cables **106**, charging cable **112**, J-Box **108**, charging station **110**, or plugged-in PEV. As will be become more clear from the description that follows, the EVSEPs **102** also transition rapidly back and forth between the ON and STANDBY, as source and load conditions change, allowing the MCUs **202** in the EVSEPs **102** to dynamically adjust, individually and collectively, the maximum allowable charging currents their respective charging stations **110** can draw.
(49) An EVSEP **102** will be configured in the “ON” state only if: 1) the PEV is properly plugged into the associated EVSEP's **102**'s charging station (as verified, for example, by a proximity check performed by and between the charging station's charging cable handle **114** and PEV), and 2) the EVSEP's MCU **202** has completed negotiating and setting a maximum allowable charging current for the PEV. Once those criteria are satisfied and the PEV commences charging, the EVSEP's MCU **202** continuously monitors the charging current based on measurements taken by the EVSEP's **102**'s Hall effect sensors **210**. As charging progresses, the EVSEP's MCU **202** communicates with the MCUs **202** in all other EVSEPs in the EV charging system **100** that are distributing charging currents to their respective charging stations **110**, via the EVSEP communications bus **116**, and repeatedly and continuously calculates and recalculates the amount of EVSEP panel **104** supply current that is available for supply to the EVSEP's associated charging station **110**, as will be described in more detail below.
(50) FIG. **5**B shows the configuration of an EVSEP **102** when in the OFF state. An EVSEP **102** will transition to the OFF state in the unlikely and unfortunate event of a fault (e.g., short circuit) occurring in the EVSEP's load circuit, or if for some reason the EVSEP **102** is unable to adjust its charge control settings in a manner that allows charging current to flow to its connected PEV. These anomalies will be detected or determined by the EVSEP's MCU **202**, with the aid of the EVSEPs Hall effect sensors **210**, and upon being detected or determined the EVSEP's MCU **202** will direct the driver circuit **206** to switch the bi-directional solid-state switch **204** OFF, as soon as it is possible, and will also generate a solenoid trigger signal that triggers the solenoid in the air gap disconnect unit **208** to open the air gap **216**, as previously described.
(51) The third and final state that the EVSEPs **102** can be configured in is the “STANDBY” state

(see FIG. **5**C). The STANDBY state may be thought of as the default state, since it is the state EVSEPs **102** remain in while waiting for PEVs to connect to their associated charging stations **110**. It is also the state that an EVSEP **102** temporarily transitions to after a PEV has already commenced charging, upon the EVSEP **102** determining that it is necessary or desired to dynamically adjust its charge control and/or circuit protection settings. When a PEV plugs into one of the charging stations **110**, the EVSEP **102** associated with that charging station **110** verifies that the PEV is properly plugged in (for example, as verified by a proximity check performed between the charging station's charging cable handle **114** and PEV's charge controller). After the PEV has properly plugged in, the PEV requests a desired charging current I_REQ from the EVSEP **102**, according to a control pilot procedure that may or may not be standardized. The EVSEP's MCU **202** then either honors the request or negotiates a lower charging current if, for example, the current available for supply from the EVSEP panel **104** is insufficient to meet the request. In either event, once the EVSEP **102** and PEV have negotiated a maximum allowable charging current I_CHARGE, the EVSEP's MCU **202** then adjusts the circuit protection it provides based on the negotiated maximum allowable charging current I_CHARGE. In one embodiment of the present disclosure the MCU **202** in each EVSEP **102** is configured to adjust the EVSEP's circuit protection continuous current rating Ir based on the negotiated maximum allowable charging current I_CHARGE (for example, Ir=125% of I_CHARGE, where Ir equals long-time pick-up, or LTPU) and the bi-directional solid-state switch **204** in each EVSEP **102** is controlled to provide overcurrent protection at the LTPU current point, similar to what is depicted in FIG. **3**.
(52) From the foregoing description it should be clear that the maximum allowable charging current I_CHARGE set by an EVSEP **102** is not static. Rather, it varies and is dynamically adjusted by the EVSEP's MCU **202** over time. At any given time the current available for supply I_AVAIL to any given EVSEP **102** depends both on the total current supply I_SUPP available from the EVSEP panel **104** and the sum of all currents allocated to the EVSEPs **102** (I.sub.1+I.sub.2+I.sub.3+ . . .) in the EV charging system **100**. In other words, I_AVAIL=I_SUPP−(I.sub.1+I.sub.2+I.sub.3+ . . .). As will be explained in more detail below, the EV charging system **100** adjusts the allocated currents (I.sub.1+I.sub.2+I.sub.3+ . . .) as PEVs plug into the charging stations **100** to charge and as PEVs complete charging and unplug from the charging stations **110**. The MCUs **202** in the EVSEPs **102** communicate and coordinate with one another over the EVSEP communications bus **116** to determine how and when adjustments to the allocated currents (I.sub.1+I.sub.2+I.sub.3+ . . .) are needed or desired, and the EVSEPs **102** dynamically adjust their maximum allowable charging currents I_CHARGE in response to changes made to the allocated currents (I.sub.1+I.sub.2+I.sub.3+ . . .). Note that in most circumstances each EVSEP **102** will utilize the full amount of current allotted to it, in other words, will dynamically adjust its maximum allowable charging current I_CHARGE to match the full amount of current allocated to it (but not necessarily in all circumstances).
(53) In addition to the circuit protection provided by the bi-directional solid-state switches **204**, each of the EVSEPs **102** further includes a ground-fault circuit interrupter (GFCI) **214**. The GFCI **214** comprises a differential current transformer and GFCI sensing circuit that continuously monitors the currents flowing through the current carrying conductors L1 and L2 conductor. Any detected imbalance in the two currents is indicative of a possible ground fault (GF). Accordingly, when an imbalance is detected, the GFCI sensing circuit sends a GF detect signal to the EVSEP's **102**'s MCU **202**, which then responds as quickly as possible to transition the EVSEP **102** to the OFF state (by triggering the air gap disconnect unit **208** to disengage and open the air gap **216** and directing the driver circuit **206** to switch the bi-directional solid-state switch **204** OFF, as previously described). It is also acceptable not to open the air gap **216** and only to turn the bi-directional solid-state switch **204** off. It should be mentioned that some conventional (prior art) EVSEs are also equipped with some sort of GFCI capability. However, GFCI protection in conventional EVSEs is only operable when the charging cable is plugged into the PEV. In contrast,

the EVSEPs **102** of the present disclosure have the ability to detect and isolate ground faults both when the charging cable **112** is plugged into the PEV and when the charging cable **112** is unplugged. The additional ability to detect and respond to ground faults when the charging cable **112** is unplugged from the PEV follows from the EVSEP's **102**'s ability to detect leakage currents flowing through its bi-directional solid-state switch **204** when the EVSEP **102** is in the STANDBY state (the state the EVSEP **102** is nominally in when not electrically connected to a PEV). The additional ground fault protection provided by the EVSEP **102** is a significant safety feature since during use the charging cable **112** can weaken to the point that one of its conductors becomes exposed (e.g., due to aging, normal wear and tear, or because of being run over by a vehicle). The exposed conductor creates a shock hazard condition that the associated EVSEP's GFCI **214** has the ability to detect as a ground fault, even when the charging cable **112** is unplugged and stored on the associated charging station's **110**'s pedestal. Accordingly, once the GFCI **214** detects the fault condition, the EVSEP **102** transitions as quickly as possible from the STANDBY state to the OFF state, to prevent any further usage of the charging cable **112** and associated charging station **110**.
(54) In one embodiment of the present disclosure the EV charging system **100** and EVSEPs **102** are designed so that they are compliant with the SAE J1772 (“J Plug”) standard, support AC Level 2 charging, and utilize an AC charging connector having L1, L2, GND, and control pilot pins similar to as illustrated in FIG. **6**A. Additionally, in one embodiment of the present disclosure (see FIG. **6**B) the handles **114** of the charging cables **112** include light-emitting diodes (LED) that indicate the current status of the charging station **110** and associated EVSEP **102**: for example, one or more of: ON (green LED), STANDBY (yellow LED), and OFF (red LED), and in another embodiment of the present disclosure also or alternatively include an LED indicator that indicates the charging station **110** is currently supplying charging current to a connected PEV (or conversely, indicating that charging has completed), and an LED indicator that cautions users of a possible fault condition. It should be mentioned that whereas the connector pins in the exemplary handle **114** depicted in FIG. **6**B are designed to comply with the SAE J1772 standard, the handle **114**, EVSEPs **102** and the EV charging system **100** overall should not be construed as being restricted to any given EV charging standard, as they can be easily adapted to comply with essentially any EV charging standard. Accordingly, the present disclosure should not be construed as being restricted to any particular standard, unless specifically recited in the appended claims.
(55) While including LED indicators on the handle **114** of the charging cable **112** is useful, one or more of the LED indicators can be alternatively (or additionally) mounted in or on pedestals or other equipment at or near the charging stations **110**. Electronic displays can also be mounted in or on the charging stations pedestals (or in or on other nearby equipment), and in one embodiment of the present disclosure the CRM **212** in the EVSEPs **102** includes computer program instructions that provide the EVSEPs' MCUs **202** the ability to estimate charging times for PEVs that plug into the charging stations **110**, for example, based on the charging currents the EVSEPs **102** supply and battery pack capacities of the PEVs, and display in real time (i.e., as the PEVs charge) estimated times to complete charging on the electronic display.
(56) The exemplary embodiments of the present disclosure described and illustrated above facilitate AC charging. In other embodiments of the present disclosure the EV charging system **100** is equipped with EVSEPs **102** that have been adapted to supply both AC and DC charging currents, depending on the PEV's needs, in which case an AC and fast DC “combo” connector having a handle/plug with pins similar to that depicted in FIG. **7** is employed. In other embodiments of the present disclosure, the EVSEPs are configured to supply only DC currents, in which case a DC-only type connector may be used.
(57) Referring now to FIGS. **8**A-**8**D, there is shown a flowchart illustrating a charge control and circuit protection method **800** that one of the EVSEPs **102** performs, in accordance with one embodiment of the present disclosure. The exemplary method **800** illustrates and highlights the dynamic circuit protection and dynamic charge control functions the EVSEP **102** performs while

operating in the context of the EV charging system **100**. At the “START” of the method **800** it is assumed that a PEV has already properly connected to the EVSEP's **102**'s charging station **110** (e.g., as confirmed by passing a J1772 proximity check performed by and between the charging cable handle **114** and PEV charge controller) and further assumed that the EVSEP **102** is in the STANDBY state and ready to supply charging current to the charging station **110** and connected PEV.

(58) At first step **802** (see FIG. **8**A), after detecting that the PEV has connected to the charging station **110**, the associated EVSEP **102** begins transmitting a control pilot signal to the PEV's charge controller, via the EVSEP panel gateway **118** and through the associated charging station **110** and charging cable **112**. For example, if the SAE J1772 standard is being followed, the control pilot signal is a +12 VDC signal, and the MCU **202** of the EVSEP **102** determines that the PEV has connected by detecting and measuring a voltage drop across a resistor divider formed between the connected handle **114** and PEV charge controller. The PEV charge controller responds to the control pilot with a charging current request I_REQ. For example, if the SAE J1772 standard is being followed, the PEV charge controller changes a resistance of one of the resistors in the charge controller, resulting in an additional voltage drop detected by the EVSEP **102** indicative of the charging current request I_REQ. At step **804**, the EVSEP's MCU **202** detects the charging current request I_REQ. Then, at step **806** the EVSEP's **102**'s MCU **202** calculates the current available for supply I_AVAIL from EVSEP panel **104** to the EVSEP **102**, i.e., I_AVAIL=I_SUPP−(I.sub.1+I.sub.2+I.sub.3+ . . .), as previously described. Next, at decision **808**, the EVSEP's **102**'s MCU **202** determines whether the current available for supply I_AVAIL is sufficient to fulfill the PEV's charging current request I_REQ. If the MCU **202** determines that I_AVAIL I_REQ, in other words that sufficient supply current is available (“YES” at decision **808**), the method **800** continues at step **810** (see FIG. **8**B). At step **810** the EVSEP **102** then honors the PEV's charging current request and sets the maximum allowable charging current to: I_CHARGE=I_REQ. Then, based on the I_CHARGE=I_REQ, at step **812** the EVSEP **102** adjusts the circuit protection settings of its bi-directional solid-state switch **204** accordingly. On the other hand, if the MCU **202** determines that I_AVAIL<I_REQ, in other words that insufficient supply current is available (“NO” at decision **808**), the method **800** branches to step **838** to determine an I_CHARGE<I_REQ and the circuit protection settings for the EVSEP's bi-directional solid-state switch **204** are set accordingly (see FIG. **8**D, described in detail below).

(59) Assuming that the EVSEP **102** honors the PEV's charging request at step **810** and has set the maximum allowable charging current to I_CHARGE=I_REQ at step **812**, before transitioning from the STANDBY state to the ON state and commencing charging, at decision **814** the EVSEP's GFCI **214** performs a ground fault check. If the ground fault test fails (“NO” at decision **814**), the method **800** ends and the PEV is not permitted to commence charging. On the other hand, if the ground fault test passes (“YES” at decision **814**), indicative of no ground fault present, at step **816** the EVSEP **102** transitions from the STANDBY state to the ON state and begins supplying current to the PEV, according to I_CHARGE=I_REQ and the circuit protection settings of the bi-directional solid-state switch **204** previously set by the EVSEP **102** based on I_CHARGE=I_REQ.

(60) As the PEV charges, the MCU **202** of the associated EVSEP **102** communicates with the MCUs **202** in the other EVSEPs **102** in the EV charging system **100**, over the EVSEP communications bus **116**, and constantly monitors and continually recalculates the available supply current I_AVAIL. So long as there is no change in I_AVAIL (“NO” at decision **818**), the PEV continues charging according to the previously set I_CHARGE and according to the circuit protection settings previously set by the EVSEP **102**, as indicated by step **820**. As the PEV continues charging, the EVSEP **102** also continues to monitor the charging progress, as indicated by decision **822**. If the EVSEP's MCU **202** determines, either by itself or in response to a “charging complete” notification from the PEV, that the PEV has completed charging (“YES” at decision **822**), the EVSEP **102** transitions to the STANDBY state and the method **800** ends. However, if the

EVSEP's MCU **202** determines that charging has not completed ("NO" at decision **822**), at step **826** the EVSEP's MCU **202** recalculates the current available for supply I_AVAIL once again, and at decision **818** queries again whether an adjustment to the maximum allowable charging current I_CHARGE is required. (It should be mentioned that, although not reflected precisely in the flowchart, the calculation in step **826** is preferably, though not necessarily, performed repeatedly and continuously in the background as the EVSEP **102** operates, from the time the PEV plugs into the charging station **110** until charging is completed.) If the EVSEP's MCU **202** ascertains no change in I_AVAIL ("NO" at decision **818**), at step **820** charging is allowed to continue according to the previously set maximum allowable charging current I_CHARGE and previously-set circuit protection settings. However, if the EVSEP's MCU **202** determines that I_AVAIL has in fact changed ("YES" at decision **818**), the method **800** continues at decision **826** in FIG. **8**C.

(61) There are a number of reasons why the current available for supply I_AVAIL to the EVSEP **102** might change. For example, I_AVAIL may decrease due to another PEV plugging into one of the charging stations **110** or may increase due to another PEV completing charging and unplugging from its charging station **110**. I_AVAIL may also decrease or increase due to an increase or decrease in the available supply current fed to the EVSEP panel **104** or in response to a charge distribution rule imposed by a rules-based charge allocation algorithm (described in more detail below). Accordingly, if the EVSEP's MCU **202** determines that I_AVAIL has in fact changed ("YES" at decision **818**), the method **800** continues in FIG. **8**C and at decision **826** the EVSEP's MCU **202** then determines whether I_AVAIL has increased or decreased. If the EVSEP's MCU **202** determines that the current available for supply I_AVAIL to the EVSEP **102** has increased ("↑" at decision **826**), the method **800** continues at step **828**. On the other hand, if the EVSEP's MCU **202** determines that the current available for supply I_AVAIL has decreased ("↓" at decision **826**), the method **800** continues at step **838** in FIG. **8**D.

(62) Assuming that the EVSEP's MCU **202** has determined that the current available for supply I_AVAIL has increased ("↑" at decision **826**), the EVSEP's MCU **202** determines whether the maximum allowable charging current I_CHARGE was previously already set to I_CHARGE=I_REQ (at step **810** above) or was set to a value less than I_REQ, i.e., I_CHARGE<I_REQ, due to the EVSEP's MCU **202** determining at decision **808** that insufficient current was then available to honor the PEV's charging current request I_REQ. If the EVSEP's MCU **202** determines that I_CHARGE is already set to I_REQ, i.e., that I_CHARGE=I_REQ ("YES" at decision **830**), the EVSEP **102** then branches back to step **820** in the flowchart (see FIG. **8**B) with the EVSEP **102** configured according to the previously set allocated charge I_CHARGE=I_REQ and previously set circuit protection settings. However, if at decision **830** the EVSEP's MCU **202** determines that I_CHARGE was set to a value less than I_REQ, i.e., was set to I_CHARGE<I_REQ due to the EVSEP's MCU **202** determining at decision **810** that insufficient current was then available to honor the PEV's charging current request I_REQ (see steps FIG. **8**D below), at step **832** the EVSEP's MCU **202** increases the maximum allowable charging current from I_CHARGE<I_REQ to I_CHARGE=I_REQ (or to the maximum charging current available), and at step **834** readjusts the circuit protection settings of its bi-directional solid-state switch **204** accordingly. After the EVSEP's **102**'s maximum allowable charging current has been readjusted to I_CHARGE=I_REQ and its circuit protection settings have been readjusted, at step **836** the EVSEP **102** transitions back to the ON state. Then the method **800** branches back to step **820** in the flowchart (see FIG. **8**A) with the EVSEP **102** configured to operate according to the updated (readjusted) maximum allowable charging current I_CHARGE=I_REQ and updated (readjusted) circuit protection settings.

(63) If the EVSEP **102** has determined at decision **826** that the current available for supply I_AVAIL to the EVSEP **102** has decreased, rather than increased (as discussed in reference to FIG. **8**B) or the EVSEP's MCU **202** determines that insufficient current is available to honor the PEV's charging current request I_REQ at decision **808** (during the time the PEV first plugs into the

charging station **110**), the method **800** continues at step **838** in FIG. **8**D, after which, at step **840**, specifically, the EVSEP's MCU **202** reduces the maximum allowable charging current from I_CHARGE to I_CHARGE′ (I_CHARGE′<I_CHARGE≤I_AVAIL) and readjusts its circuit protection settings based on the magnitude of the new maximum allowable charging current I_CHARGE′ at step **842**. After the EVSEP's **102**'s maximum allowable charging current I_CHARGE′ and circuit protection settings have been readjusted, the method **800** branches back to step **820** in the flowchart (see FIG. **8**B) with the EVSEP **102** configured to operate according to the updated (readjusted) maximum allowable charging current I_CHARGE′ and updated (readjusted) circuit protection settings. The method **800** then continues in the manner described above, until the EVSEP's MCU **202** determines at decision **822** that the PEV has completed charging ("YES" at decision **822**), after which the method **800** ends.

(64) It should be emphasized that the various steps and decisions in the exemplary method **800** described above are not necessarily performed in the order shown in the flowchart. Some steps and decisions in the method **800** are, or may be, performed constantly, continuously, or simultaneously, e.g. in the background, as the method **800** is performed, rather than as a chronological sequence of events. For example, for safety reasons the GFCI check at decision **814** is preferably performed constantly, even when a PEV is not plugged into the EVSEP's **102**'s associated charging station **100**, and, as was mentioned above, the calculation in step **826** is preferably performed repeatedly and continuously in the background by the EVSEP's MCU **202** as the EVSEP **102** supplies current to its associated charging station **110** and as source and load conditions in the EV charging system **100** change.

(65) It is also important to point out that the exemplary method **800** is an illustration of how just a single one of the EVSEPs **102** in the EV charging system **100** dynamically adjusts its charge control and circuit protection functions as source and load conditions change over time. In a preferred embodiment of the present disclosure, all of the other EVSEPs **102** in the EV charging system that are distributing charging currents to their respective charging stations **110** perform substantially the same method **800**. In one embodiment of the present disclosure the plurality of EVSEPs **102** are configured to operate collectively, with their respective MCUs **202** communicating with one another and exchanging source and load information over the EVSEP communications bus **116** in real time, to dynamically coordinate an allocation (or apportionment) of the total supply current I_SUPP available from the EVSEP panel **104** among the EVSEPs **102**. As the EV charging system **100** operates, the MCUs **202** in the EVSEPs **102** repeatedly and continuously monitor the sum of all allocated currents (I.sub.1+I.sub.2+I.sub.3+ . . .), so that the total supply current I_SUPPL available from the EVSEP panel **104** is never exceeded. Whenever the MCUs **202** determine that insufficient current is available from the EVSEP panel **104** to satisfy the charging current request I_REQ of any or more charging stations **110** and associated PEVs, the allocated currents are reallocated or reapportioned and the MCUs **102** in the one or more of the EVSEPs **102** lower their maximum allowable charging currents, from I_CHARGE to I_CHARGE′, similar to as described above, so that the total current drawn from by the EVSEPs **102** does not exceed the total current supply I_SUPP available from the EVSEP panel **104**. The EVSEPs **102** then also readjust their circuit protection setting based on the reduced maximum allowable charging currents I_CHARGE. This collective and coordinated dynamic charge allocation and dynamic circuit protection process is performed continuously by the EVSEPs **102** as the EV charging system **100** operates. By apportioning the total supply current I_SUPP available from the EVSEP panel **104** among the EVSEPs **102**, all PEV are able to continue charging, albeit with one or more PEVs charging at a lower charging current I_REQ than requested. Apportioning and allocating also allows an additional PEV to connect to the EV charging system **100** and charge. Although the charging current allocated to it via the associated EVSEP **102** and charging station **110** will also be less than the charging current it requests, the additional PEV will nevertheless be able to charge, whereas if no apportionment was made the additional PEV would have to wait for

other PEVs to complete charging before it could commence charging.
(66) There are various ways the MCUs **202** in the EVSEPs **102** can be programmed to allocate or apportion the total supply current I_SUPP among the EVSEPs **102**. In one embodiment of the present disclosure, the total supply current I_SUPPL is simply allocated equally among the EVSEPs **102**/charging stations **110**. In other words, the total supply current I_SUPP is divided by the number of charging stations **110** in use and each EVSEP **102**. In another embodiment of the present disclosure, a weighting-based algorithm is employed, according to which the MCUs **202** in the EVSEPs **102** track and monitor the time each of the PEVs has been charging, receive battery information from the individual PEVs, including their individual battery capacities, and estimate the time it will take for each of the PEVs to complete charging at a given charging current. Weighing these factors, the EVSEPs **102** then apportion the total supply current I_SUPP among the EVSEPs **102** similar to as illustrated in FIG. **9**. In another embodiment of the present disclosure, some PEVs are given priority over others, for example, in response to a PEV user indicating to the EV charging system **100** a willingness to pay a higher price for charging. FIG. **10** illustrates for example, how the weighting-based algorithm depicted in FIG. **9** can be modified to include this priority-based dependency. Whichever algorithm is employed, each time the EV charging system **100** makes an adjustment to the allocated currents (I.sub.1+I.sub.2+I.sub.3+ . . .), one or more of the EVSEPs **102** then adjust its/their maximum allowable charging current I_CHARGE and also preferably, though not necessarily, readjusts its/their circuit protection settings based on the new value of I_CHARGE. The EVSEPs **102** perform these adjustments dynamically and in real time as the EV charging system **100** operates. Note that the circuit protection adjustments made in the examples presented in FIGS. **9** and **10** involve adjustments to the continuous current ratings Ir of the EVSEPs **102**. The continuous current ratings Ir are calculated and set and recalculated and reset dynamically by the EVSEPs' MCUs **202** as currents to the EVSEPs in the EV charging system **100** are allocated and reallocated. For example, as illustrated in FIG. **9**, after the current allocated to Vehicle A has been lowered from 50 A to 20 A (due to Vehicle B plugging into charge), the EVSEP associated with Vehicle A then lowers its continuous current rating Ir from 62.5 A (125% of 50 A) to 25 A (125% of 20 A). Although not indicated in FIGS. **9** and **10**, those of ordinary skill in the art will appreciate and understand from the teachings of this disclosure that other circuit protection attributes of the EVSEPs, besides just the continuous current ratings Ir, may also (or alternatively) be dynamically adjusted as the EV charging system **100** operates. For example, the long-time delay, short-time delay, and/or instantaneous pickup settings of any one or more of the EVSEPs can also (or alternatively) be adjusted if desired or as conditions warrant.
(67) It should be mentioned that the weight-based charge allocation algorithms depicted in FIGS. **9** and **10** are just two examples of how the EVSEPs **102** can be configured to operate when the total current supply I_SUPPL available from the EVSEP panel **104** may be insufficient to satisfy the charging current requests of PEVs. For example, in one embodiment of the present disclosure, instead of apportioning the total current supply I_SUPPL among the EVSEPs **102** and allowing PEVs to charge all at the same time, the MCUs **202** in the EVSEPs **102** are programmed so that the EVSEPs **102** are allowed to charge, perhaps at their full requested charging currents I_REQ, according to a time-slotted schedule, whereby one or more of the EVSEPs **102** supplying charging currents temporarily transition(s) to the STANDBY state while one or more of the other EVSEPs **102** are given priority and allowed to continue supplying charging current to their respective charging stations **110** and connect PEVs. Once the PEV(s) associated with the priority EVSEP(s) **102** has/have completed charging in its/their scheduled time slots, the remaining EVSEPs **102** that had previously transitioned to the STANDBY state transition back to the ON state to complete the charging of their PEVs, for example, in one or more subsequent time slots.
(68) It should also be mentioned that the dynamic charge control and dynamic circuit protection capabilities of the EV charging system, including any one of the rules-based charging allocation algorithms discussed above, can be designed to extend across multiple EVSEP panels that are

configured in a more expansive EV charging system network. All that would be required to implement the more expansive EV charging system would be to construct communications links between the various EVSEP panels. In this more expansive EV charging system the total current supply I_SUPPL available at any one of the EVSEP panels could be increased or decreased as desired or necessary, even as the various EVSEPs **102** dynamically coordinate the allocation or apportionment of charging currents to the various EVSEPs **102**. A dynamic allocation scheme, similar to described above, could also be implemented at the panel level.

(69) The ability of the EVSEPs **102** to individually and collectively dynamically adjust their maximum allowable charging currents and circuit protection settings is beneficial since it avoids uneven usage of the EV charging system infrastructure. The more even usage avoids the one-size-fits-all approach of conventional EV charging systems, which requires use of oversized power distribution equipment, yet still allows most every PEV charging need to be accommodated. Another advantage the EV charging system **100** has over conventional EV charging systems is that the dynamic charge control and dynamic circuit protection functions are performed locally, with the EVSEPs' MCUs' **102** communicating, cooperating, and coordinating with one another over the EVSEP communications bus **116**. No centralized computer is required and no cloud connection or network is required, thus avoiding a centralized control scheme that would undesirably pose as a single point of failure and make the EV charging system vulnerable to hackers and cyber attacks. Yet another advantage the EV charging system **100** has over conventional EV charging systems is that the frame sizes of the EVSEPs **102** are configurable. This attribute is afforded by the EVSEPs **102** use of bi-directional solid-state switches **204** to provide short-circuit and overcurrent protection. This frame-size configurability is not possible in prior art EV charging systems, which rely on electromechanical circuit breakers, external to the EV charging system, for circuit protection.

(70) The exemplary EV charging system **100** and EVSEPs **102** described above are well-suited for AC PEV charging, in particular, SAE J1772 AC Level 1 (120 VAC, 16 A Max) and/or AC Level 2 (208-240 VAC, 80 A Max) PEV charging. This AC charging configuration is illustrated in FIG. **11**. In another embodiment of the present disclosure, the EVSEPs **102** are modified so that they are capable of supporting DC-fast charging, for example, SAE J1772 DC Level 1 (50V to 1 kV, 80 A Max) and/or DC Level 2 (50V to 1 kV, 1000 A Max) PEV charging, as illustrated in FIG. **12**. To support DC charging, in one embodiment of the present disclosure each of the EVSEPs **102** is modified to further include an AC/DC converter, for example, a three-phase AC-to-DC bridge rectifier **1202** (as shown in FIG. **12**) in the case of a three-phase AC charging system. The three-phase AC-to-DC bridge rectifier **1202** is configured to mate with a three-phase implementation of the bi-directional solid-state switch **204** (similar to that depicted in FIG. **4**). The modified EVSEP **102** further includes a DC/DC converter **1204** that steps the converted DC voltage down to a level suitable for charging. All of the other functions and capabilities of the AC version of the EVSEPs **102** described above, including the dynamically adjustable circuit protection charge control capabilities are included in the modified (i.e., DC-fast charging) version of the EVSEP **102** depicted in FIG. **12**.

(71) In one embodiment of the present disclosure the bi-directional solid-state switches **204** in the DC-modified versions of the EVSEPs **102** are not only configured and controlled by their respective MCUs **202** to provide dynamically adjustable circuit protection, the MOSFETs in the bi-directional solid-state switches **204** are further configured and controlled to serve as a front end for power factor correction. In yet another embodiment of the present disclosure, the bi-directional solid-state switches **204** are also (or alternatively) configured to regulate the DC voltage produced at the output of the three-phase AC-to-DC bridge rectifier **1202** or output of the DC/DC converter **1204**.

(72) FIG. **13** is a drawing depicting an EV charging system **1300**, according to another embodiment of the present disclosure. The EV charging system **1300** is particularly well-suited to serve as a

fleet charging facility, for example, for EV buses used by a municipality. The EV charging system **1300** comprises an overhead busway **1302** (e.g., Starline 1200 A, 4-pole) that distributes AC power supplied from two or three phases of the secondary winding of a three-phase utility transformer **1304** to a plurality of EVSEPs **102** configured along the overhead busway **1302**. Because of the high currents (up to and possibly above 1000 A) that can be supplied by secondary of the utility transformer **1304**, the EV charging system's main service is preferably distributed underground, for example, with each of the conductors, L1, L2, L3 (if three-phase) routed in separate PVC conduits/pipes (e.g., 4″, schedule 40 PVC conduits). Instead of configuring and housing the EVSEPs **102** within a distribution panel, the EVSEPs **102** are configured as busplugs that connect, i.e., “plug into” matching receptacles configured at various locations in the overhead busway **1302**. Additional charging stations can be added by simply plugging additional EVSEPs/busplugs **102** into vacant receptacles and attaching corresponding charging cables **1306** to the busplugs along the busway **1302**. The EV charging system **1300** supports, AC and DC charging, by simply plugging in one of the EVSEPs **102** like that used in FIG. **11** (AC charging) or the EVSEP **102** depicted in FIG. **12**, which has been modified to include three-phase AC-to-DC bridge rectifier **1202** and DC/DC converter **1204** (DC-fast charging).

(73) In the EV charging system **1300** described above, DC-fast charging is supported by modifying the EVSEPs **102** to include a three-phase AC-to-DC bridge rectifier **1202** and DC/DC converter **1204**. In an alternative embodiment, a single, large AC/DC converter is employed to convert the large line-to-line AC voltages (e.g., 480 VAC) at the utility transformer **1304** secondary, thus obviating the need to perform the AC/DC conversion at the outputs of the EVSEPs **102**. With that modification made, the EV charging system **1300** becomes a DC-fast-charging-only facility.

(74) Additional embodiments that follow are based upon a difference in requirements between an Underwriters Laboratory (UL) panelboard and a UL 2594 EVSE. Since the UL 67 panelboard with UL breakers must be safely operated in all applications, solid-state circuit breakers (SSCBs) in an EVSE specific application allows for reduced equipment costs an increased safety along with other optimizations not possible in other applications. Moreover, due to the inherent nature of SSCBs that includes millisecond reaction time to stop the flow of current, SSCBs are employable in EVSE applications that are not easily attainable with other methods of power delivery.

(75) FIG. **14** is a drawing of an EVSEP **102-1** that includes an internal ground fault current transformer (GFCT) **214-1** that is configured to detect ground fault currents in two-phase conductors (L1, L2) and the neutral conductor N, according to an embodiment of the present disclosure. The EVSEP **102-1** has a first bi-directional solid-state switch **204-1** and a first current sensor **210-1** coupled in series between an L1 line-side terminal **218-1** and an L1 load-side terminal **220-1**. The EVSEP **102-1** has a second bi-directional solid-state switch **204-2** and a second current sensor **210-2** coupled in series between an L2 line-side terminal **218-2** and an L2 load-side terminal **220-2**. A neutral conductor N is coupled between a neutral line-side terminal **218-3** and a neutral load-side terminal **220-3**, wherein a portion of the neutral conductor N is in current sensing proximity of the GFCT **214-1**. A first microcontroller unit (MCU) **202-1** is configured to receive current sensing signals from the first current sensor **210-1** and the second current sensor **210-2**. The first MCU **202-1** is further configured to receive a ground fault sense signal from the GFCT **214-1** that indicates a ground fault on the neutral conductor N and/or the two-phase conductors (L1, L2). The first MCU **202-1** is further configured to receive and/or transmit data over the EVSEP communication bus **116**. As such the first MCU **2201** is configured to provide individual and independent control of the of the N number of EV charging stations over the EVSEP communication bus **116**. The EVSEP **201-1** further includes a first driver **206-1** that is configured to drive the first bi-directional solid-state switch **204-1** and the second bi-directional solid-state **204-2** between ON in a state of conduction and OFF in a state of non-conduction.

(76) FIG. **15** is a drawing of a version the EVSEP **102-1** that is configured to respond to a ground fault sense signal generated by an external version of the GFCT **214-1**. In this embodiment, the

GFCT **214-1** is configured to detect ground fault currents in two-phase conductors (L1, L2) and the neutral conductor N. However, in this embodiment, the GFCT **214-1** is external to the EVSEP **102-1** and the neutral conductor N is coupled between a neutral bar and the neutral load-side terminal **220-3**. A portion of the neutral conductor N is in current sensing proximity of the GFCT **214-1** to ensure generation of the ground fault sense signal whenever a ground fault event involving the neutral conductor N occurs.

(77) FIG. **16** is a drawing of an embodiment of the EVSEP **102-1** that includes a version of the internal GFCT **214-1** that is configured to detect ground fault currents in the three-phase conductors (L1, L2, L3) and the neutral conductor N. In this embodiment, the EVSEP **102-1** further includes a third bi-directional solid-state switch **204-3** and a third current sensor **210-3** coupled in series between an L3 line-side terminal **218-4** and an L3 load-side terminal **220-4**. The MCU **202-1** is configured to receive a current sense signal from the third current sensor **210-3**.

(78) FIG. **17** is a drawing of a version the EVSEP **102-1** that is configured to respond to the ground fault sense signal generated by the external version of the GFCT **214-1**. In this embodiment, the GFCT **214-1** is configured to detect ground fault currents in three-phase conductors (L1, L2, L3) and the neutral conductor N. However, in this embodiment, the GFCT **214-1** is external to the EVSEP **102-1** and the neutral conductor N is coupled between the neutral bar and the neutral load-side terminal **220-3**. A portion of the neutral conductor N is in current sensing proximity of the GFCT **214-1** to ensure generation of the ground fault sense signal whenever a ground fault event involving the neutral conductor N occurs.

(79) Other embodiments of the EVSEP **102-1** depicted in FIGS. **14** through **17**, and referring back to FIG. **2**, may further include the closeable air gap **216** of the air gap disconnect unit **208**, between input (line-side) terminals **218** and output (load-side) terminals **220**. These other embodiments of the EVSEP **201** are configurable in three possible states that are as follows: the ON state during which the closable air gap **216** of the air gap disconnect unit **208** of associated ones of the N number of charging stations is closed and the bidirectional solid-state switch **204** of the individual active ones of an N number of charging stations is ON; the STANDBY state during which the closable air gap **216** of the air gap disconnect unit **208** of associated ones of the N number of charging stations is closed and the bidirectional solid-state switch **204** of the individual active ones of the N number of charging stations is OFF; and the OFF state during which the closable air gap **216** of the air gap disconnect unit **208** of associated ones of the N number of charging stations is open and the bidirectional solid-state switch **204** of the individual ones of the N number of charging stations is OFF.

(80) FIG. **18** is a drawing of an exemplary version of the EV charging system **100** that is configured to charge two electric vehicles per each of a first EVSEP **102-1** and a second EVSEP **102-2** that make up the EVSEP panel **104**. In this exemplary embodiment, the first EVSEP **102-1** has a first internal GFCT **214-1** and a second EVSEP **102-2** has a second internal GFCT **214-2** as depicted in FIG. **14**, and further includes a gateway and pilot module **222** configured to communicate with pedestals **224-1**, **224-2**, and **224-3**. In this embodiment, the pedestal **224-1** is a dual pedestal that is configured to charge two PEVs.

(81) In operation, the phase L1 provides charging current flowing through a first bi-directional solid-state switch **204-1** and the neutral N to a first PEV using the dual pedestal **224-1**. The phase L2 provides charging current flowing through a second bi-directional solid-state switch **204-2** and the neutral N to a second PEV using the dual pedestal **224-1**. Ground faults associated with the dual pedestal **224-1** are sensed by the first GFCT **214-1** that in response generates a ground fault sense signal that is received by a first MCU **202-1** which in turn commands a first driver **206-1** to shut off the flow of current flowing through the affected phase(s) L1 and/or L2. Pilot signals from the PVEs charging from the dual pedestal **224-1** is received by the gateway+pilot module **222**, which in turn routes information associated with the pilot signal to the first MCU **202-1** by way of the communications bus **116**.

(82) A PEV using the single pedestal **224-2** is charged by a charging current flowing through phase L1 controlled by a third bi-directional solid-state switch **204-3** that comprises a second EVSEP **102-2**. The charging current provided to the associated PEV returns to the busbar by way of the neutral N. Ground faults associated with the second pedestal **224-2** are sensed by a second GFCT **214-2** that in response generates a ground fault sense signal that is received by a second MCU **202-2**, which in turn commands a second driver **206-2** to shut off the flow of current flowing through the affected phase(s) L1 and/or L2. A pilot signal from the PVE charging from the single pedestal **224-2** is received by the gateway+pilot module **222**, which in turn routes information associated with the pilot signal to the second MCU **202-2** by way of the communications bus **116**.
(83) A PEV using the single pedestal **224-3** is charged by a charging current flowing through phase L2 controlled by a fourth bi-directional solid-state switch **204-4** that comprises the second EVSEP **102-2**. The charging current provided to the PEV charging at the single pedestal **224-3** returns to the busbar by way of the neutral N. Ground faults associated with the third pedestal **224-3** are sensed by a second GFCT **214-2** that in response generates a ground fault sense signal that is received by the second MCU **202-2** which in turn commands a second driver **206-1** to shut off the flow of current flowing through the affected phase L2. A pilot signal from the PVE charging from the single pedestal **224-3** is received by the gateway+pilot module **222**, which in turn routes information associated with the pilot signal to the second MCU **202-2** by way of the communications bus **116**. It should be noted that the pilot module can be separate from the gateway without deviating from the scope of the disclosure.
(84) The GFCT **214-2**, in FIG. **18**, detects ground fault currents in both EV systems connected to the dual-pedestal **224-1**. The system, in response to a ground fault, will shut off the flow of current through both L1 and L2. The system can restart L1, test for ground fault and enable L1 if no ground fault is present. If a ground fault is present in L1 the system will shut off the flow of current through L1 and turn L2 on. This will isolate the fault to only the impacted EVSE.
(85) FIG. **19** is a drawing of a version of the electric vehicle charging system **100** that is configured to charge four electric vehicles per EVSEP panel **104** and includes a gateway and power over ethernet (PoE) module **226** to communicate with smart pilot modules **228-1**, **228-2**, and **228-3** in pedestals **224-1**, **224-2**, and **224-3**, respectively. It is to be understood that the PoE module **226** is not restricted to a particular type and may be various other types of substantially low-voltage DC and data modules.
(86) The smart pilot modules **228-1**, **228-2**, and **228-3** are configured to provide high-level communication with PEVs that are to be charged or are being charged. The smart pilot modules **228-1**, **228-2**, and **228-3** are customizable to adapt to future communication protocols or other user interfaces that might be required. For example, if “plug and charge” is allowed by PEVs, the smart pilot modules **228-1**, **228-2**, and **228-3** are configured to translate the vehicle data of the PEVs and report back to the EVSEP panel **104** to initiate charging sessions. The smart pilot modules **228-1**, **228-2**, and **228-3** may also be configured to be used with a radio frequency identification (RFID) or credit card reader to send the payment data back to the EVSEP panel **104** for session activation. The EVSEP panel **104** is typically the control hub of the EV charging system **100** and there is no low level power at the pedestals **224-1**, **224-2**, and **224-3**. As such, the smart pilot modules **228-1**, **228-2**, and **228-3** are configured to operate at a safe low voltage like 48 VDC PoE, for example. In this regard, power over ethernet is supplied over category **5** or better (CAT PoE) cabling that couples the gateway+PoE module **226** with the smart pilot modules **228-1**, **228-2**, and **228-3**. In some embodiments, the smart pilot modules **228-1**, **228-2**, and **228-3** are configured to be updated remotely via over the air updates through the EVSEP panel **104**. Moreover, the smart pilot modules **228-1**, **228-2**, and **228-3** are configured to be replaced easily if a new EV standard is released without needing to replace the EVSEP panel **104**. This configuration makes the EV charging system **100** easily upgradable and future proof.
(87) FIG. **20** is a drawing of a version of the electric vehicle charging system **100** that is configured

to charge two electric vehicles per the first EVSEP **102-1** and the second EVSEP **102-2**. The second EVSEP **102-2** is configured to respond to ground fault sense signals generated by external GFCTs **214-2** and **214-3** located at the single pedestals **224-2** and **224-3**, respectively. During operation, ground faults associated with the second single pedestal **224-2** are detected by the first external GFCT **214-2**, which in response generates a ground fault sense signal that is processed by the smart pilot module **228-2**. Information indicating a ground fault is passed immediately over the CAT PoE to the gateway+Power Over Ethernet Module **226**, which in turn passes the ground fault indication to the second MCU **202-2**. In response to the ground fault indication, the second MCU **202-2** commands the second driver **206-2** to stop the flow of current through the appropriate one(s) of the third bi-direction switch **204-3** and/or the fourth bi-direction switch **204-4**.

(88) Ground faults associated with the third single pedestal **224-3** are detected by the second external GFCT **214-3**, which in response generates a ground fault sense signal that is processed by the third smart pilot module **228-3**. Information indicating a ground fault is passed immediately over the CAT PoE to the gateway+Power Over Ethernet Module **226**, which in turn passes the ground fault indication to the second MCU **202-2**. In response to the ground fault indication, the second MCU **202-2** commands the second driver **206-2** to stop the flow of current through the fourth bi-direction solid-state switch **204-4**.

(89) FIG. **21** is a drawing of a version of the electric vehicle charging system **100** that is configured to charge three electric vehicles per the first EVSEP **102-1** and the second EVSEP **102-2**. Addition of a third phase L3 to the EVSEP panel **104** provides for charging six electric vehicles in total. A fifth bi-directional solid-state switch **204-5** is coupled in series with a third phase L3 conductor between the bus bar and a tri-pedestal version of the first pedestal **224-1**. The fifth-bi-directional switch semi-conductor **204-5** is integrated with the first bi-directional solid-state switch **204-1** and the second bi-directional solid-state switch **204-2** within the first EVSEP **102-1**. The first MCU **201-1** is configured to command the first driver **206-1** to control current flowing through the fifth bidirectional solid-state switch **204-5** and thereby control the charging current to a third electric vehicle charging at the first pedestal **224-1**. In this embodiment, the GFCT **214-1** is configured to detect ground fault currents in three-phase conductors (L1, L2, L3) and the neutral conductor N that are associated with the first pedestal **224-1**.

(90) A sixth bi-directional solid-state switch **204-6** is coupled in series with another third phase L3 conductor between the bus bar and the first single pedestal **224-2**. The sixth-bi-directional solid-state switch **204-6** is integrated with the third bi-directional solid-state switch **204-3** and the fourth bi-directional solid-state switch **204-4** within the second EVSEP **102-2**. The second MCU **202-2** is configured to command the second driver **206-2** to control current flowing through the sixth bidirectional solid-state switch **204-6** and thereby control the charging current to a sixth electric vehicle charging at a fourth single pedestal **224-4**. In this embodiment, the second GFCT **214-2** is configured to detect ground fault currents in three-phase conductors (L1, L2, L3) and the neutral conductor N associated with the pedestals **224-2**, **224-3**, and **224-4**.

(91) FIG. **22** is a drawing of a version of the electric vehicle charging system **100** that is configured to charge three electric vehicles per the first EVSEP **102-1** and the second EVSEP **102-2**. The second EVSP **102-2** is configured to respond to ground fault sense signals generated by external GFCTs **214-2**, **214-3**, and **213-4** located at single pedestals **224-2**, **224-3** and **224-4**, respectively. In exemplary embodiments, one or more bi-directional power semiconductors such as SiC MOSFETs may be used to realize the bi-directional solid-state switches **204-1**, **204-2**, **204-3** **204-4**, **204-5**, and **204-6**. Also, the GFCTs **214-1** through **214-4** may be the GFCI **214** depicted in FIG. **2**.

(92) FIG. **23** is a drawing of an electromechanical version the EVSEP **102-1** that is configured to respond to a ground fault sense signal generated by an external version of the GFCT **214-1** that is configured to detect ground fault currents in two-phase conductors. This exemplary embodiment does not employ solid state switches. Instead, the electromechanical version the EVSEP **102-1** employs the air gap disconnect unit **208-1** having the closeable air gap **216-1**. The air gap

disconnect unit **208-1** is coupled between corresponding ones of the line-side terminals **218-1**, **218-2**, and **218-4** and the load-side terminals **220-1**, **220-2**, and **220-4**. In this exemplary embodiment, the air gap disconnect unit **208-1** can by either of the relay or the contactor type.
(93) In operation, the GFCT **214-1**, detects ground fault currents in any associated EV systems (not shown in FIG. **23**). In response to a ground fault, the MCU **202-1** will shut off the flow of current through L1, L2, and L3. The MCU **202-1** is configured to then restart L1, test for ground fault and enable L1 if no ground fault is present. If a ground fault is present in L1 the MCU **202-1** will shut off the flow of current through L1 and turn L2 on. The MCU **202-1** will then test for ground fault and enable L2 if no ground fault is present. If a ground fault is present in L2 the MCU **202-1** will shut off the flow of current through L2 and turn L3 on. The MCU **202-1** will next test for ground fault and enable L3 if no ground fault is present. If a ground fault is present in L3 the MCU **202-1** will shut off the flow of current through L3. This will isolate the fault to only the impacted phase of the phases L1, L2, and L3.
(94) FIG. **24** is a drawing of an electromechanical version of the electric vehicle charging system **100** that is configured to charge two electric vehicles per EVSEP including the gateway and pilot module **222** configured to communicate with pedestals **224-1** and **224-2**, according to an embodiment of the present disclosure. This exemplary embodiment does not employ solid state switches. Instead, the electromechanical version the EVSEP **102-1** employs the air gap disconnect unit **208-1** having the closeable air gap **216-1**. Similarly, the EVSEP **102-2** employs the air gap disconnect unit **208-2** having the closeable air gap **216-2**.
(95) While various embodiments of the present disclosure have been described, they have been presented by way of example and not limitation. It will be apparent to persons skilled in the relevant art that various changes in form and detail may be made to the exemplary embodiments without departing from the true spirit and scope of the present disclosure. Accordingly, the scope of the present disclosure should not be limited by the specifics of the exemplary embodiments but, instead, should be determined by the appended claims, including the full scope of equivalents to which such claims are entitled.
(96) See Appendix A for additional description of the disclosed smart circuit system.
(97) It is contemplated that any of the foregoing aspects, and/or various separate aspects and features as described herein, may be combined for additional advantage. Any of the various embodiments as disclosed herein may be combined with one or more other disclosed embodiments unless indicated to the contrary herein.
(98) Those skilled in the art will recognize improvements and modifications to the preferred embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A smart circuit system configured to directly interface with a circuit breaker, the system comprising: an interface for communication with the circuit breaker; a ground fault circuit interrupter (GFCI) for detecting fault conditions and activating the circuit breaker to isolate the fault condition; a load control device for managing power supply based on the status of a pilot signal; an electric vehicle supply equipment (EVSE) that is activated when the pilot signal is present; a diagnosis module configured to evaluate the status and health of the connected circuit breaker, by comparing the input voltage of the smart circuit with voltages measured by other smart circuits fed from the same voltage source within the same panel, detecting substantial voltage deviations and voltage trends over time to identify potential circuit breaker degradation; and a communication module for establishing a network with other smart circuits within the same panel for data exchange, with or without a gateway, enabling each smart circuit to perform local calculations and make independent decisions based on the shared data.

2. The smart circuit system of claim 1, wherein the system is configured to be connected within an existing panel or integrated as part of any other system where load control or EVSE functionality is required.
3. The smart circuit system of claim 1, wherein the GFCI is configured to be externally added to the circuit breaker.
4. The smart circuit system of claim 1, wherein the communication module is configured to communicate over a mesh network for data exchange among the smart circuits within the panel.
5. The smart circuit system of claim 1, wherein the diagnosis module is configured to use a threshold voltage difference to identify a nominally operating system and track increasing deviations from this threshold to detect potential circuit breaker degradation.
6. The smart circuit system of claim 1, wherein the shared data among the smart circuits include the health status of each smart circuit and the voltage readings at the input of each smart circuit.
7. The smart circuit system of claim 1, further comprising a load monitoring module configured to measure current flowing into each load, determining the load type, and assessing the health status of the load.
8. A method for enhancing circuit protection using the smart circuit system of claim 1, comprising the steps of detecting fault conditions, activating the circuit breaker, controlling power supply, metering power consumption, diagnosing circuit breaker health, and trends of the deviations over time to identify potential circuit breaker degradation, and establishing communication among smart circuits within the same panel.
9. The method of claim 8, further comprising the steps of externally adding a GFCI function to a circuit breaker and monitoring current flow into each load for load type identification and health assessment.
10. The method of claim 8, wherein the diagnosing step involves comparing input voltage with voltages measured by other smart circuits within the same panel, detecting significant deviations, and trends of the deviations over time to identify potential circuit breaker degradation.
11. An electric vehicle (EV) charging system according to claim 1 comprising: two or more EV charging stations; a plurality of the electric vehicle supply equipment (EVSE) units, each of which comprises a microcontroller unit configured to provide individual and independent control of the of the two or more EV charging stations by controlling one or two bi-directional solid-state switches and a driver configured to switch the one or two bi-directional switches to control a maximum allowable charging current available to the two or more EV charging stations when one or more plug-in vehicle (PEVs) are plugged into associated ones of the two or more EV charging stations and provide circuit protection to the associated ones of the two or more EV charging stations and the one or more PEVs; one or two phase conductors coupled between load-side terminals of the charging stations and corresponding ones of the one or two bi-directional solid-state switches; and a neutral conductor coupled to a neutral terminal of the two or more EV charging stations, wherein the neutral conductor is configured to complete charging circuits with associated ones of the one or two phase conductors supplying charging current to the one or more PEVs.